MUHAMMAD MEMON

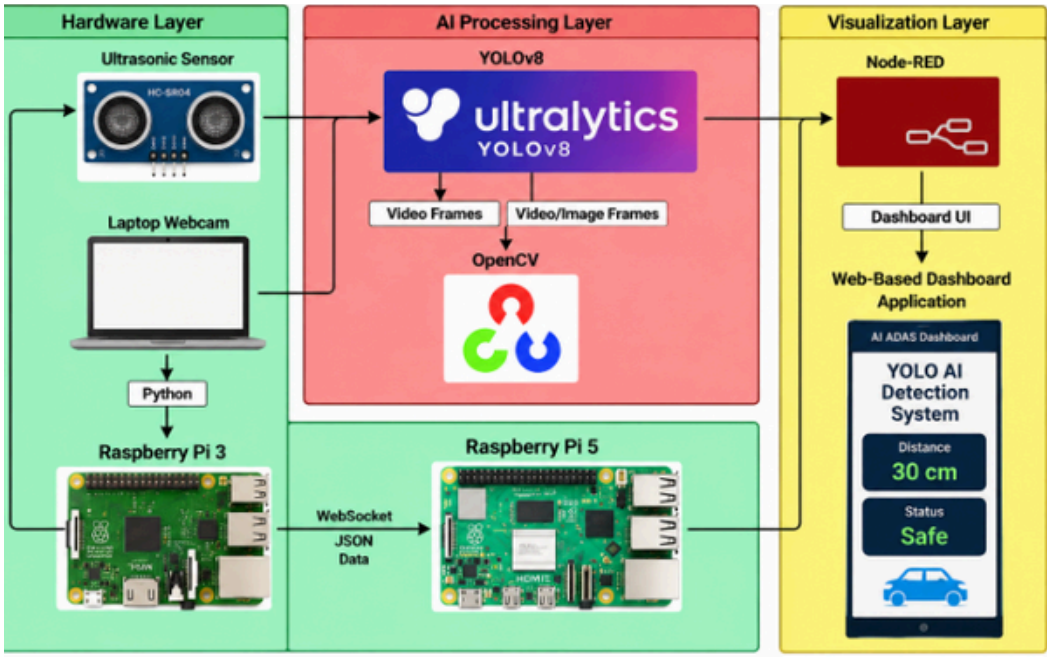
momemon2003@gmail.com
linkedin.com/in/muhammadmemon
(089) 986 - 3355
https://github.com/MMemon2003

COMPUTER ENGINEERING AT TECHNOLGICAL UNIVERSITY DUBLIN

4TH YEAR FINAL PROJECT - AI ADAS DETECTION SYSTEM

PROJECT OVERVIEW

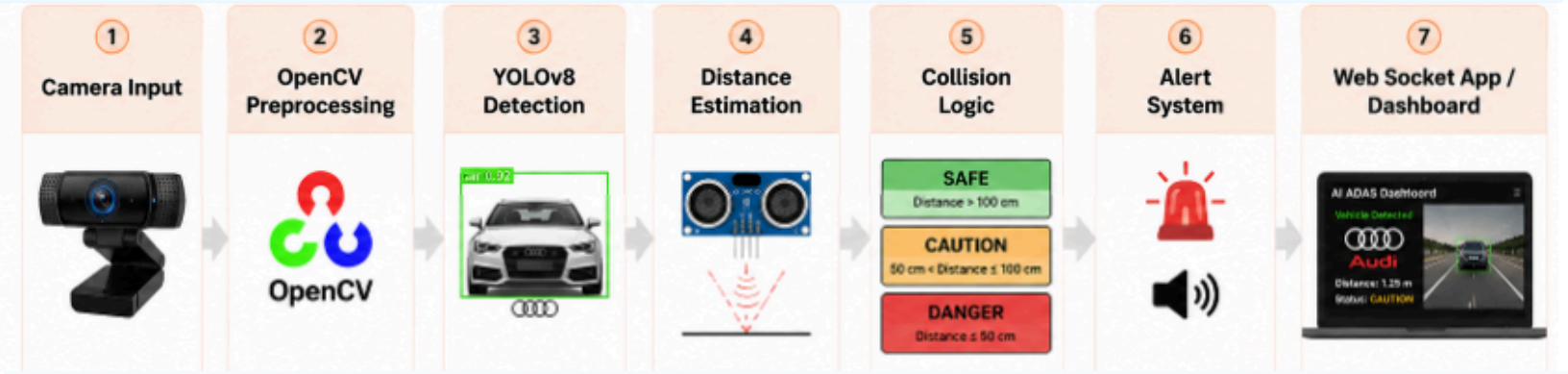
A prototype driver-assistance system that detects road users, estimates their distance and issues collision warnings using AI on low-cost embedded hardware



What?

- Developed a low-cost Artificial Intelligence (AI) based Advanced Driver Assistance System (ADAS) to improve driver awareness during reversing, blind-spot monitoring, and parking manoeuvres.
- The system combines embedded AI, computer vision, distance sensing, and IoT technologies to detect nearby vehicles, recognise vehicle logos, estimate obstacle distance, and generate collision warnings.
- Designed as an affordable alternative to commercial ADAS systems using Raspberry Pi embedded hardware.
- Provides real-time dashboard visualisation and mobile notifications through Node-RED and WebSocket communication

SYSTEM WORKFLOW



Seven-stage process from camera input to the WebSocket dashboard

1. Camera Input	A Pi Camera V3 or USB webcam captures live video frames of the surrounding environment.
2. OpenCV Preprocessing	OpenCV processes and resizes image frames to prepare them for AI-based object detection.
3. YOLOv8 Detection	The YOLOv8n model identifies road users and target objects, including vehicles and pedestrians, in the camera feed.
4. Distance Estimation	An HC-SR04 ultrasonic sensor measures the distance to nearby objects to support proximity monitoring.
5. Collision Logic	The system compares measured distances against configured thresholds to classify proximity as SAFE, CAUTION or DANGER.
6. Alert System	Visual and audio alerts notify the user when an object enters a configured caution or danger zone.
7. WebSocket App / Dashboard	Node-RED and WebSocket communication display detection results, distance readings and warning status on a dashboard accessible over the local network.

MUHAMMAD MEMON

COMPUTER ENGINEERING AT TECHNOLGICAL UNIVERSITY DUBLIN

momemon2003@gmail.com
linkedin.com/in/muhammadmemon
(089) 986 - 3355
https://github.com/MMemon2003

HARDWARE COMPONENTS

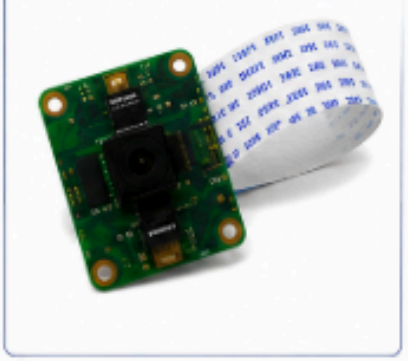
1 Raspberry Pi 3



2 Raspberry Pi 5



3 Raspberry Pi Camera Module V3



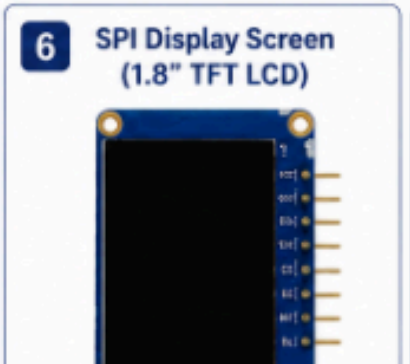
4 Laptop Webcam



5 HC-SR04 Ultrasonic Sensor



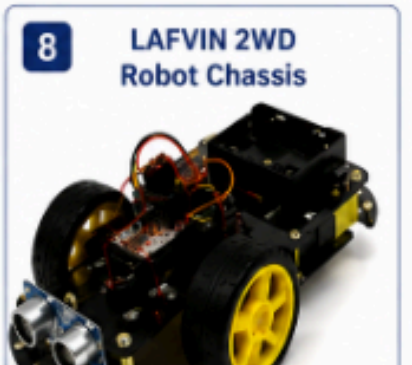
6 SPI Display Screen (1.8" TFT LCD)



7 L298N Motor Driver



8 LAFVIN 2WD Robot Chassis



Raspberry Pi 3

What?

- Raspberry Pi 3 is a single-board computer used for embedded applications and system control.

How?

- Runs the core software of the AI ADAS system.
- Executes Python scripts, handles sensor data, performs preprocessing, and sends data to Raspberry Pi 5.

Raspberry Pi 5

What?

- Raspberry Pi 5 is a high-performance single-board computer with improved speed and connectivity.

How?

- Handles AI inference using the YOLOv8 model.
- Receives data (frames and sensor information) from Pi 3 and processes it for object detection.

Raspberry Pi Camera Module V3

What?

- Raspberry Pi Camera Module V3 is an official high-quality camera module designed for Raspberry Pi.

How?

- Captures real-time video frames.
- Provides the video input to Raspberry Pi 3 for further processing.

Laptop Webcam

What?

- A USB webcam used as an alternative video input source.

How?

- Captures live video when the Raspberry Pi Camera Module is not available.
- The video feed is sent to Raspberry Pi 3 for processing.

HC-SR04 Ultrasonic Sensor

What?

- HC-SR04 is an ultrasonic distance sensor that uses sound waves to measure distance.

How?

- Emits ultrasonic pulses and measures the time taken for the echo to return.
- Calculates the distance to nearby obstacles for collision detection and warning.

SPI Display Screen (1.8" TFT LCD)

What?

- A 1.8-inch SPI TFT LCD display that communicates with the Raspberry Pi.

How?

- Displays real-time information such as detected objects (e.g., Audi logo), distance, system status, and warnings.
- Shows the output received from Raspberry Pi 5 via the SPI protocol.

L298N Motor Driver

What?

- L298N is a dual H-bridge motor driver module for controlling DC motors.

How?

- Controls the speed and direction of the motors.
- Receives control signals from Raspberry Pi 3 to drive the robot.

LAFVIN 2WD Robot

What?

- A 2-wheel drive robotic platform for DIY and educational projects.

How?

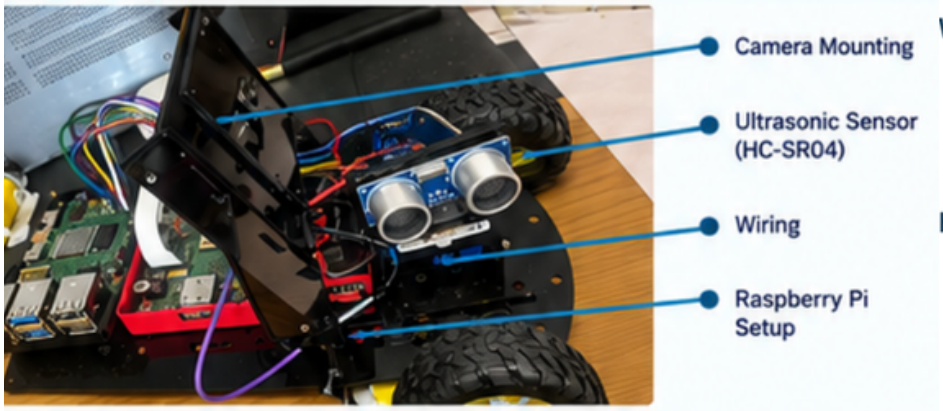
- Provides the physical base and mobility for the system.
- Integrates all hardware components including motors, sensors, and controllers into a complete ADAS prototype.

MUHAMMAD MEMON

COMPUTER ENGINEERING AT TECHNOLGICAL UNIVERSITY DUBLIN

momemon2003@gmail.com
linkedin.com/in/muhammadmemon
(089) 986 - 3355
https://github.com/MMemon2003

AI DETECTION AND SOFTWARE CONFIGURATION



Prototype hardware: Raspberry Pi, camera and ultrasonic sensor on a 2WD robotic car.

What?

- The system uses a camera feed and computer vision to identify vehicles pedestrians and vehicle logos.
- YOLOv8n is used as a lightweight object-detection model so inference can run on embedded hardware.

How?

- OpenCV resizes frames and supports brightness processing/augmentation before detection.
- YOLOv8n was fine-tuned by transfer learning on the LOGO-17 dataset.
- The workload is split across two boards: Raspberry Pi 3 runs detection and distance sensing, while Raspberry Pi 5 hosts the Node-RED dashboard.

Results

- Live detection examples show vehicles and people marked with bounding boxes.
- An example detection payload contains a distance reading of 50 cm and a detected car with confidence 0.87.

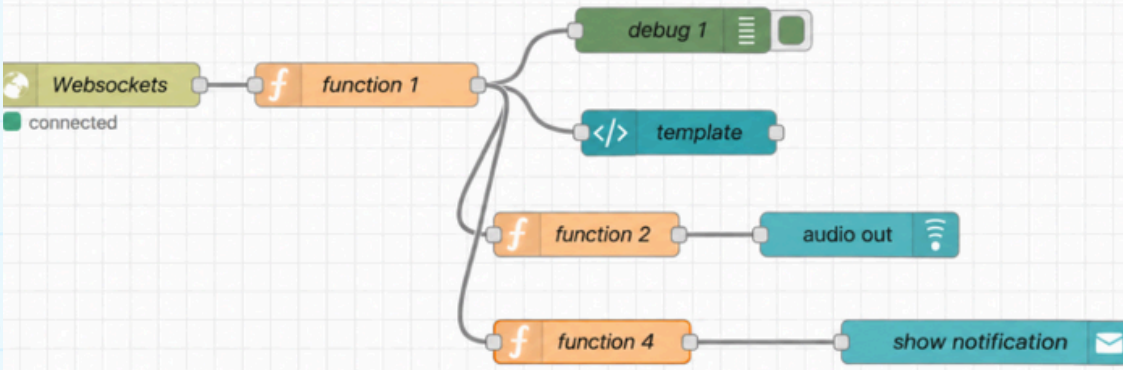
NODE-RED CONFIGURATION AND WEBSOCKET DATA FLOW

Receive Data
Detection results and distance readings arrive from the Raspberry Pi over WebSockets.

Process Messages
Incoming payloads are parsed into detection, distance and alert fields.

Update Dashboard
Processed values are sent to the Node-RED dashboard for live display (web and mobile).

Trigger Alerts
Visual and audio alerts are activated when configured distance thresholds are met (SAFE, CAUTION, DANGER).




```
"distance": 50,
"detections": [
  {
    "label": "car",
    "confidence": 0.87
  }
]
```



What?

Node-RED receives detection and distance data from the Raspberry Pi over WebSocket communication. It processes incoming messages and updates the dashboard in real time.

How?

- The WebSocket input connects to a function node, which processes the incoming payload.
- The processed message is sent to the debug and template nodes and to separate function branches for audio output and notifications.
- The dashboard displays detection information, distance readings and warning status on the local network.

Results

- The Node-RED flow provides the dashboard output and supports visual and audio alerts when configured distance thresholds are met.

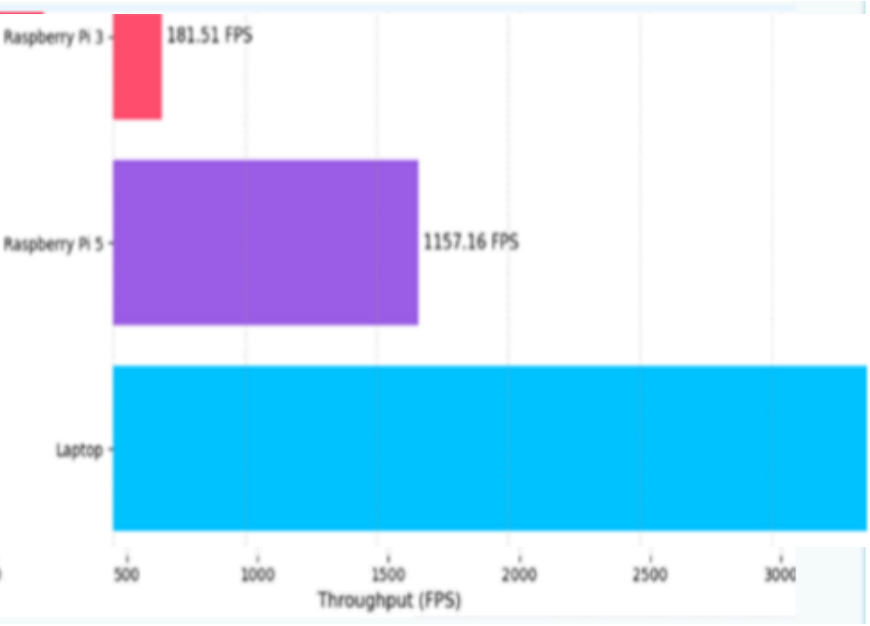
MUHAMMAD MEMON

COMPUTER ENGINEERING AT TECHNOLGICAL UNIVERSITY DUBLIN

momemon2003@gmail.com
linkedin.com/in/muhammadmemon
(089) 986 - 3355
https://github.com/MMemon2003

RESULTS, TESTING AND PERFORMANCE

181.5 FPS	1,157.2 FPS	2,861.9 FPS
Raspberry Pi 3	Raspberry Pi 5	Laptop



What?

The project benchmarked processing throughput on Raspberry Pi 3, Raspberry Pi 5 and a laptop.

Results

- Pi 3: 181.5 FPS.
- Pi 5: 1,157.2 FPS.
- Laptop: 2,861.9 FPS.
- Pi 5 is the strongest embedded platform of the two Raspberry Pi boards.

The reported figures show higher throughput on the laptop and Raspberry Pi 5 than on Raspberry Pi 3.

DASHBOARD OUTPUT AND SYSTEM VALIDATION



Mobile dashboard view (accessible on phone)

Testing and outcome

- The project poster reports that all eight system tests passed.
- Cars and people were detected live with bounding boxes.
- Detection results, distance information and alerts reached the dashboard and phone.
- The dashboard displays the detected object and distance reading, with warning status available to the user.

Detects vehicles, pedestrians and vehicle logos using a lightweight YOLOv8n model.

Estimates distance with an HC-SR04 ultrasonic sensor and uses distance thresholds to trigger warnings.

Displays detections, distance information and alerts through a Node-RED dashboard accessible on a phone over a local network.

Runs fully on embedded hardware (edge AI) without relying on the cloud.